R PROGRAMMING

Data Cleaning & Missing Data

Experiment Report

Name	Tanisha Gupta
Roll Number	23102A0002

This experiment explores practical techniques for identifying, correcting, and validating data quality issues using R. It covers invalid numerical values, safe calculations, performance comparison, and strategies for handling missing data.

1. Aim and Objectives

The aim is to apply data-cleaning techniques in R and assess the quality of datasets before further analysis.

• Identify invalid, extreme, and missing values.

• Use conditional statements, loops, and functions to clean data.

• Handle calculation warnings and errors safely.

• Compare loop-based and vectorized operations.

• Apply appropriate strategies for different types of missing data.

• Validate and export the cleaned datasets.

2. Tools and Packages

dplyr supports data manipulation, **naniar** summarizes and visualizes missingness, **skimr** provides descriptive summaries, and **microbenchmark** compares execution times.

3. Cleaning Blood Pressure Data

The heart-disease dataset is used to demonstrate how to handle incorrect resting blood-pressure values. The data includes examples of negative values, missing values, and unusually high values. A custom function checks each value and applies the relevant rule.

Data condition	Action
Missing value	Retained as NA
Negative value	Replaced with NA
Value greater than 250	Capped at 250
Other values	Kept unchanged

The cleaned values are stored separately from the original data. Safe calculation functions are then used to calculate the mean and the cholesterol-to-blood-pressure ratio while accounting for missing values and invalid operations.

4. Loop and Vectorized Comparison

Outlier detection is implemented using both a loop and a vectorized logical expression. The results are compared for consistency, and execution time is measured. In the recorded run, vectorized processing was faster than the loop-based approach, although timings depend on the operation, data size, and system.

Method	Mean time (µs)	Median time (µs)
Loop-based	71.63	67.04
Vectorized	4.60	4.27

5. Validation Results

The cleaned blood-pressure column was checked for missing values, negative values, and values exceeding the chosen upper limit.

Measure	Result
Missing values in cleaned blood pressure	10
Minimum	94
Maximum	250
Mean	133.97
Median	130
Negative values remaining	No
Values above 250 remaining	No

No negative values or values above 250 remained after cleaning. Missing values were retained where appropriate rather than being filled automatically.

6. Handling Missing Data

A census-income dataset is used to demonstrate different data-quality issues, including NA, NaN, blank strings, and an impossible age value. The experiment converts the impossible age to NA, replaces blank workclass values with “Unknown,” removes rows with NaN in hours worked per week, and fills missing ages using the median.

7. Final Validation

Missingness summaries, a missing-data visualization, and descriptive statistics are used to assess the cleaned data. The checks show that the targeted age and blank-workclass issues were handled. Some missing values remain in other columns, so the dataset is not completely free of missing data.

8. Conclusion

The experiment demonstrates how R can be used to identify and handle invalid data, apply conditional cleaning rules, protect calculations with error handling, and compare loop-based with vectorized operations. It also illustrates multiple strategies for missing data, followed by validation and export. These steps help improve data quality before conducting further analysis.